

ECZEMA

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INTRODUCTION

Eczema or dermatitis is skin inflammation caused by internal or external stimuli. It tends to be a chronic skin condition affecting patients' quality of life. While this condition is common, challenges frequently encountered with diagnosing and treating dermatitis in primary care include:

- Variability in its presentation across different ages, durations, and ethnic groups.
- Unfamiliarity with the multitude of treatment options available.
- Complexity of patients' Ideas, Concerns and Expectations (ICE) regarding their skin condition and its treatment.

With appropriate management, many patients can be successfully managed in the primary care setting, while those who would benefit from specialist evaluation and treatment should be referred early.

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

Eczema is a clinical diagnosis characterized by the following features¹:

History

- Itching or pruritus is a cardinal symptom.
- Certain triggers such as contactants may occasionally cause an acute onset of symptoms.

Physical Examination

- Lesion morphology may vary by duration and intensity of inflammation:
- Acute dermatitis suggests an acute trigger: subcutaneous oedema, vesicles, erosions, weeping / oozing and crusting.
- Chronic dermatitis suggests a lower-potency trigger occurring repeatedly over time: thickened plaques, post-inflammatory hyperpigmentation, or hypopigmentation, lichenification, prurigo nodules.
- Subacute dermatitis lies between acute and chronic dermatitis and may have features of both.
- Lesions which have superimposed bacterial infection (impetiginisation) may display weeping, yellow crusting or signs of deeper infection such as erythema and warmth.
- Considerable variations are seen in lesion presentation:
- It may be more difficult to appreciate erythema in patients with darker skin.
- In the Asian context, post-inflammatory hyperpigmentation is more common than hypopigmentation.
- The 'flexural' distribution is characteristic of atopic dermatitis. However, other causes of dermatitis may have different distribution patterns.

- Lesion morphology is often modified by prior treatment or patient scratching the skin lesions.

Causes of Dermatitis

Atopic Dermatitis

- Chronic, relapsing-remitting skin condition which typically starts in infancy or childhood.
- The exact cause of atopic dermatitis is unknown, but it is believed that genes, the immune system, and the environment play a role in the disease.
- Exacerbated by various triggers:
 - Environmental: extremes of temperature, low humidity, house dust mites, contact allergens.
 - Irritants: perspiration, insect bites, contactants.
 - Infections: skin infections e.g. *Staphylococcus aureus*, systemic infections.
 - Food: trigger in small minority of patients, particularly children with severe or refractory disease.
 - Often associated a personal and/or family history of atopic disorders such as asthma, allergic rhinitis and/or allergic conjunctivitis.
- Distribution varies with age of onset:
 - Infantile: face, scalp, neck, extensor areas of limbs and trunk.
 - Childhood and adolescence: elbow, knee, wrist and neck flexures, periorbital and perioral regions of face.
 - Adult: lesions become more subacute to chronic, may develop lichenification or chronic papular lesions due to habitual rubbing or scratching.
- Other characteristic atopic features:
 - Keratosis pilaris
 - Hyperlinear palms

- Ichthyosis vulgaris
- Atypical vascular responses – mid-facial pallor, white dermographism
- Complications of atopic dermatitis include:
- Bacterial superinfection
- Eczema herpeticum – suspect when there is a sudden eruption of vesicles on erosions in a patient with underlying atopic dermatitis. Frequently associated with fever, malaise, and lymphadenopathy.

Contact Dermatitis

- Dermatitis due to exogenous substances, with typical sources including:
 - Household
 - Occupational
 - Iatrogenic
 - Traditional medications: ‘Hong You’, ‘Qing Cao You’, ‘Tiger Balm’ *etc.*
 - Topical medications: cleansing or disinfectant agents
- Types of contact dermatitis:
 - Irritant Contact Dermatitis (ICD): local toxic effect due to topical substance or physical insult
 - Allergic Contact Dermatitis (ACD): immunologically-mediated hypersensitivity reaction to a substance to which the patient may have been previously sensitised to
- Additional points in the history to take note of:
 - The patient’s occupation and handedness

- Use of any Personal Protective Equipment (PPE)
- Prior attempts to treat other skin diseases. Often inappropriate self-treatment with traditional or over-the-counter topicals may precipitate contact dermatitis superimposed on the existing skin disease.
- Please note that work-related contact dermatitis may be an Occupational Skin Disease which is reportable to the Ministry of Manpower under the Workplace Safety and Health Act.

Stasis Dermatitis

- Dermatitis changes are typically present on the dependent areas of the lower limbs.
- Patients may present with a history of chronic venous insufficiency (CVI) characterized by the following:
 - Dependent oedema
 - Vascular changes: telangiectasias, varicose veins
 - Skin changes: pigmentation, shiny, waxy taut skin (lipodermatosclerosis)
 - Venous ulceration: typically a shallow, irregular ulcer over the medial malleolar region with a fibrinous base.
- As patients may not be aware of their symptoms or signs of CVI, stasis dermatitis may be the first presentation of CVI in many patients in the polyclinic setting.

Asteatotic / Xerotic Eczema

- Occurs in areas of dry skin, particularly on extensor surfaces of forearms and shins in older adults.
- Affected skin is dry and may have a pattern of cracks and superficial vertical and horizontal fissures ('crazy-paving').
- Despite being associated with the winter months in temperate climates, this form of eczema can occur any time of the year in Singapore.

Discoid / Nummular Eczema

- Characterised by markedly pruritic, coin-shaped often subacute dermatitic plaques which tend to occur in the upper and lower limbs. When acutely inflamed, the lesions may be extremely itchy, oozing and crusted.
- May occur in the absence of any trigger.

Seborrhoeic Dermatitis

- Occurs in infants or adults, characterised by erythematous plaques with greasy scaling typically affecting skin with higher density of sebaceous glands - scalp, eyebrows, ears and nasolabial folds.
- May be associated with HIV infection or neurological disorders e.g. Parkinson's disease.

Dyshidrotic Eczema / Pompholyx / Hand Feet Vesicular Eczema

- A type of hand and feet dermatitis characterised by the formation of firm vesicles on the fingers and soles.
- May be intensely pruritic and relapsing, flares may be triggered by stress and contact dermatitis.

Lichen Simplex Chronicus / Prurigo Nodularis

- Lichen Simplex Chronicus: circumscribed area of lichenified dermatitis due to repeated rubbing or scratching.
- Prurigo nodularis: nodular lesions due to chronic picking and scratching due to chronic pruritus. May be due to underlying atopic dermatitis or other causes of chronic pruritus.

MANAGEMENT OF ECZEMA

The components of treatment are:

1. Identify and address triggers
2. Improving the epidermal barrier
3. Treat skin inflammation
4. Treat pruritus
5. Use of antimicrobials where appropriate
6. Patient education

1. Identify and Address Triggers

- Based on the clinical diagnosis, screen for and address possible triggers:

Diagnosis	Possible Triggers	Strategies to Address Triggers
Atopic dermatitis	• Environmental: extremes of temperature, low humidity, house dust mites, contact allergens • Irritants: perspiration, insect bites, contactants • Infections: skin infections e.g. <i>Staphylococcus aureus</i>, systemic infections • Food: trigger in small minority of patients, most commonly children with moderate to	• House dust mite avoidance advice (see Appendix A). • Treat any superimposed bacterial infection where present. • Consider referrals for allergy testing in appropriate patients.

Diagnosis	Possible Triggers	Strategies to Address Triggers
	severe atopic dermatitis not responding to initial treatment.	
Contact dermatitis	• Screen for household, occupational and iatrogenic contactants. • In particular, consider asking direct questions about topical treatment with traditional or Western medications.	• Eliminate suspected contactants if possible or recommend PPE. • Consider referrals for allergy testing in appropriate patients.
Stasis dermatitis	Risk factors for chronic venous insufficiency: • Obesity • Prolonged standing • History of deep vein thrombosis	• Elevate lower limbs at rest. • Compression stockings. • Referral to vascular surgery where appropriate.
Asteatotic / xerotic eczema	Dry / cold weather	• Avoid excessively warm baths; lukewarm or cold water is preferred. • Intensive use of emollients.

2. Improving the Epidermal Barrier

- Most patients with dermatitis have dry skin (xerosis). This should be treated by:
- Using mild soap-free or moisturising cleansers.
- Avoiding hot showers/baths of excessive duration. Lukewarm or cold water is preferred.
- Regular use of moisturisers:
- Occlusives form a barrier on the skin surface to prevent moisture loss.
- Humectants bind water within the skin to replace lost moisture.
- If there is excessive oozing or weeping, use compresses with astringent agents to dry up excessive fluid.

Type	Drug	Dose	Cost	Remarks
Occlusive	Emulsifying Ointment	1 application OM/BD	\$	May be used as a moisturiser May also be used as a soap
Occlusive	White Soft Paraffin	1 application BD/TDS	\$	May be used on dry or chapped lips
Humectant	Aqueous Cream	1 application BD/TDS	\$	May also be used as soap substitute, cleanser, and makeup remover
Humectant	Urea cream	1 application BD/TDS	\$\$	Keratolytic effect may be useful for scaly lesions Caution: May cause mild stinging, itching and skin rashes Avoid use in children due to keratolytic effect
Astringent	Potassium permanganate	1 application as needed	\$	Dispensed as concentrated solution, dilute 1 part to 8 parts of water before use. Do not apply directly on skin.

Type	Drug	Dose	Cost	Remarks
				Moisten gauze or cotton balls and apply as compress over weepy / oozing skin. Remove after 10 to 20 minutes and allow to dry. Can be used up to twice a day. May stain skin and clothing.

3. Treat Skin Inflammation

Topical Corticosteroids (TCS)

- Topical corticosteroids (TCS) are the mainstay of treating skin inflammation.
- Select the appropriate vehicle for the type of skin condition being treated:
- Ointments are more persistent than creams and are useful for thicker, more localized lesions. Note that they tend to deliver a higher amount of topical steroid to the skin compared to a cream of the same concentration.
- Creams have a less greasy consistency and are better for applying on larger surfaces of skin.
- Select the appropriate potency of TCS for the intensity of inflammation and area of skin being treated²:

Potency	Drug(s)	Dose	Cost	Remarks
Potent	Betamethasone valerate 0.1% cream/ointment Mometasone furoate 0.1% cream	1 application OD/BD 1 application OD	\$	Betamethasone valerate 0.1% ointment is the most potent.
Moderate	Betamethasone valerate 0.025%, 0.05% cream/ointment	1 application OD/BD	\$	-

Potency	Drug(s)	Dose	Cost	Remarks
Mild	Hydrocortisone 1% cream	1 application OD/BD	\$	Recommended for neonates / infants / young children For mild lesions or lesions on the face

- Select combination drugs where appropriate. Avoid the use of combination drugs containing high-potency steroids (e.g., Combiderm, Gentrison) without clear clinical indication.

Drug	Dose	Cost	Remarks
Hydrocortisone 1% with clioquinol 3%	1 application OD/BD	\$	Clioquinol has antifungal and antibacterial activity against gram-positive bacteria
Betamethasone 0.025%/0.1% with clioquinol 3%	1 application OD/BD	\$	
Betamethasone dipropionate 0.05% and salicylic acid 3% ointment (Betacycllic)	1 application OD/BD	\$	Use limited to resistant lesions e.g. lichenified atopic dermatitis, chronic plaque psoriasis and prurigo nodularis. Once the lesions have flattened, it is recommended to switch to a topical corticosteroid of lower strength until complete resolution.

- A sufficient amount of TCS should be applied. 1 fingertip unit (FTU) is the amount of cream squeezed out from the tip of an adult finger to the volar distal interphalangeal joint (DIPJ) crease and is sufficient for 2 adult palm-sizes of skin³.

- TCS are safe if used for appropriate indications, at appropriate potencies, for appropriate durations and under medical supervision.

Adverse events are rare, but include:

- Skin atrophy and bruising
- Telangiectasias
- Acneiform eruptions
- Hypertrichosis
- Transcutaneous absorption leading to hypothalamic-pituitary-adrenal axis suppression.
- In pregnancy, the use of mild-to-moderate TCS has not been associated with adverse maternal or neonatal outcomes⁴.
- Review the patients in 1-2 weeks after initiating TCS for response. Once improved, taper the TCS dose in terms of potency or frequency. TCS can be used until resolution of the rash – this includes a tapering regime.

Topical Calcineurin Inhibitors (TCI)

- Topical calcineurin inhibitors are non-steroidal anti-inflammatory agents which are useful as adjuncts to topical corticosteroids.
- They do not have the side effect of causing skin atrophy despite prolonged use.
- They are suitable for use in sensitive skin areas such as those prone to skin atrophy such as the face, around the eyes, neck, groin or skin flexures. Potential glaucoma or cataract risk around the eyes are additional considerations.

Drug	Dose	Cost	Remarks
Pimecrolimus 1% cream	1 application OD/BD	\$\$\$ (\$57.86/tube)	For use in children 3 months or older Should not be used in pregnancy

- Adverse effects include:
 - Local symptoms during first few days of treatment: burning, pruritus, soreness, stinging. Usually self-resolving as atopic dermatitis lesions heal.
 - Skin warts have been observed with use; discontinue use if there is worsening of skin warts or if they do not respond to conventional treatment.
 - Prior to starting this medication, inform patients that the US FDA has a 'black box warning' stating 'Long-term safety of topical calcineurin inhibitors has not been established. Although a causal relationship has not been established, rare cases of malignancy (e.g. skin malignancy and lymphoma) have been reported'.
 - In general, continuous long-term use of TCI in any age group should be avoided and application limited to areas of involvement with atopic dermatitis

4. Treat Pruritus

- Sedating antihistamines may be useful to decrease itching and permit sleep during flares⁵. There is a lack of high-quality evidence on the efficacy of non-sedating antihistamines (cetirizine, loratadine)⁶.

5. Use of Antimicrobials Where Appropriate

- Signs of bacterial superinfection include weepy or impetiginized lesions. This should be treated with systemic antibiotics targeted against Gram-positive agents, in particular *Staphylococcus aureus*.
- Non-penicillin allergic: cloxacillin, cephalexin.
- Penicillin allergic: erythromycin, clindamycin.

- Topical antiseptics may be useful adjuncts

6. Patient Education (For Atopic Dermatitis)

- Importance of regular moisturizer use
- Removal of triggers, especially House Dust Mites (see [Appendix A](#))
- Appropriate use of TCS and FTU. Fear of topical corticosteroids is common and should be addressed upfront⁷.

REFERRALS

- Indications for referral to a dermatologist include:
 - There is diagnostic uncertainty.
 - The skin condition does not respond to treatment in primary care.
 - Specialist-level treatments may be required e.g. ultraviolet therapy, systemic immunosuppression, or biologic agents.
- Indications for Emergency Department referral include:
 - Suspected eczema herpeticum.
 - Clinically unwell patient e.g. febrile, toxic-appearing.
 - Extensive disease e.g. erythroderma

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APPENDIX A: HOUSE DUST MITE AVOIDANCE ADVICE

What are House Dust Mites?

- House Dust Mites are microscopic creatures that are found everywhere, especially in mattresses, pillows, bedsheets, clothes, soft toys, sofas and carpets.
- Individuals who suffer from asthma, allergic rhinitis or eczema are commonly allergic to House Dust Mite allergens.

Simple and Effective Measures

- Wash bedsheets and pillowcases in hot water that is at least 60°C weekly or fortnightly.
- Avoid beddings such as pillows and comforters that are made of natural fibres such as feathers. Replace them with items made from synthetic fibres.
- Remove stuffed toys as well as thick and heavy curtains in the bedroom that can trap dust.
- Damp dusting should be used to clean surfaces. Avoid feather dusters.
- Air-conditioners, if used, should be cleaned regularly.
- Avoid using carpets in the room.

Other Measures

- Use mite-proof mattresses, pillow covers and blankets
- Use good quality vacuum cleaners or air purifiers.

Adapted from “How to Avoid House Dust Mites” pamphlet published Jan 2021 by National University Hospital